By 2050 a ‘useless’ class might emerge not merely because of an absolute lack of jobs or lack of relevant education, but also because of insufficient mental stamina.

Obviously, most of this is just speculation. At the time of writing – early 2018 – automation has disrupted many industries but it has not resulted in massive unemployment. In fact, in many countries, such as the USA, unemployment is at a historical low. Nobody can know for sure what sort of impact machine learning and automation will have on different professions in the future, and it is extremely difficult to estimate the timetable of relevant developments, especially as they depend on political decisions and cultural traditions as much as on purely technological breakthroughs. Thus even after self-driving vehicles prove themselves safer and cheaper than human drivers, politicians and consumers might nevertheless block the change for years, perhaps decades.

However, we cannot allow ourselves to be complacent. It is dangerous just to assume that enough new jobs will appear to compensate for any losses. The fact that this has happened during previous waves of automation is absolutely no guarantee that it will happen again under the very different conditions of the twenty-first century. The potential social and political disruptions are so alarming that even if the probability of systemic mass unemployment is low, we should take it very seriously.

In the nineteenth century the Industrial Revolution created new conditions and problems that none of the existing social, economic and political models could cope with. Feudalism, monarchism and traditional religions were not adapted to managing industrial metropolises, millions of uprooted workers, or the constantly changing nature of the modern economy. Consequently humankind had to develop completely new models – liberal democracies, communist dictatorships and fascist regimes – and it took more than a century of terrible wars and revolutions to experiment with these models, separate the wheat from the chaff, and implement the best solutions. Child labour in Dickensian coal mines, the First World War and the Great Ukrainian Famine of 1932–3 constituted just a small part of the tuition fees humankind paid.

The challenge posed to humankind in the twenty-first century by infotech and biotech is arguably much bigger than the challenge posed in the previous era by steam engines, railroads and electricity. And given the immense destructive power of our civilisation, we just cannot afford more failed models, world wars and bloody revolutions. This time around, the failed models might result in nuclear wars, genetically engineered monstrosities, and a complete breakdown of the biosphere. Consequently, we have to do better than we did in confronting the Industrial Revolution.